Last week:

- Some common graphs (complete bipartite, complete t-partite).
- Operations involving graphs (complement, disjoint union, join, cartesian product).
- Multigraphs and digraphs.
- Vertex degrees (handshaking lemma); interactions with connectivity.
- r-regular graphs (all vertices have degree r).
  - If r and n are not both odd and $r \leqslant n - 1$, there exists an r-regular graph in n-vertices.
  - Any graph is contained in infinitely many regular graphs.

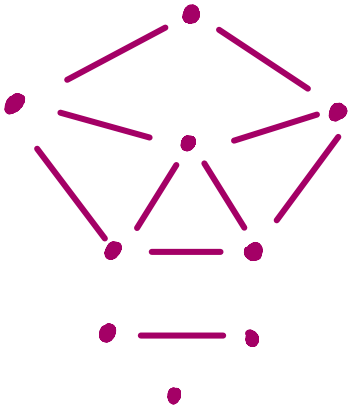
This week:

- Degree sequences.
- Graphical sequences, Havel–Hakimi theorem.
- Graph isomorphisms and automorphisms.

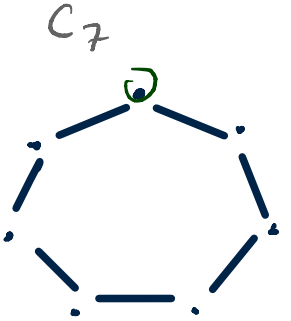
Definition. *A **degree sequence** of a graph in n-vertices is a list of its n vertex degrees (don't discard repeated degrees).*

- If $V(G) = \{v_1, v_2, \ldots, v_n\}$, then a degree sequence is $\deg v_1, \deg v_2, \ldots, \deg v_n$ (depends on the labeling of $V(G)$).
- Usual choice: arrange vertex degrees in non-decreasing list $\deg v_1 \geqslant \deg v_2 \geqslant \cdots \geqslant \deg v_n$.

Example (Some graphs and their degree sequences)**.**



S: 4, 3, 3, 3, 3, 2, 1, 1, 0



S: 2, 2, 2, 2, 2, 2, 2

Next: which sequences come from some graph?

Definition. *A finite sequence* s *of nonnegative integers is called* ***graphical*** *if it is a degree sequence of some graph* G*. In this case we say that* G *is a realization of* s*.*

Exercise. *Suppose* $s: d_1 \geqslant d_2 \geqslant \cdots \geqslant d_n$ *is graphical. Then*

- s *has an even number of odd entries.*

G has an even number of odd vertices (Corollary 2.3)

- $n-1 \geqslant d_1 \geqslant \cdots \geqslant d_n$.

In G deg v $\leq n-1$ $\forall v$

- *if* $d_1, \ldots, d_k = n-1$, *then* $d_n \geqslant k$.

We have k-vertices which are neighbors of all other vertices.

Example. *Which of the following sequences are graphical?*

(a) $s_1: 3,3,2,2,1,1$.



(b) $s_2: 6,5,5,4,3,3,3,2,2$.

5 odd entries, not graphical.

(c) $s_3: 7,6,4,4,3,3,3$.

n = 7 → problem not graphical.

(d) $s_4: 3,3,3,1$.

Have Three entries = n-1 (n=4), but $d_n = 1 < 3$ not graphical.

Theorem ((2.10) Havel–Hakimi). *A sequence* $s: d_1 \geqslant d_2 \geqslant \cdots \geqslant d_n$, *where* $d_1 \geqslant 1$, *is graphical if and only if the sequence*

$$s_1: d_2 - 1, d_3 - 1, \ldots, d_{d_1+1} - 1, d_{d_1+2}, \ldots, d_n$$

is graphical.

Delete first entry d_1, Then substract 1 from the next d_1 entries.

E.g:

S: ~~5~~, 4, 4, 3, 2, 2, 2, 1, 1

S_1: 4-1, 4-1, 3-1, 2-1, 2-1, 2, 1, 1

3, 3, 2, 1, 1, 2, 1, 1

Theorem ((2.10) Havel–Hakimi). *A sequence* $s\colon d_1 \geqslant d_2 \geqslant \cdots \geqslant d_n$, *where* $d_1 \geqslant 1$, *is graphical if and only if the sequence*

$$s_1\colon d_2 - 1, d_3 - 1, \ldots, d_{d_1+1} - 1, d_{d_1+2}, \ldots, d_n$$

is graphical.

$\Leftarrow$) If s_1 is graphical realized by G_1 with $V(G_1) = \{v_2, \ldots, v_n\}$

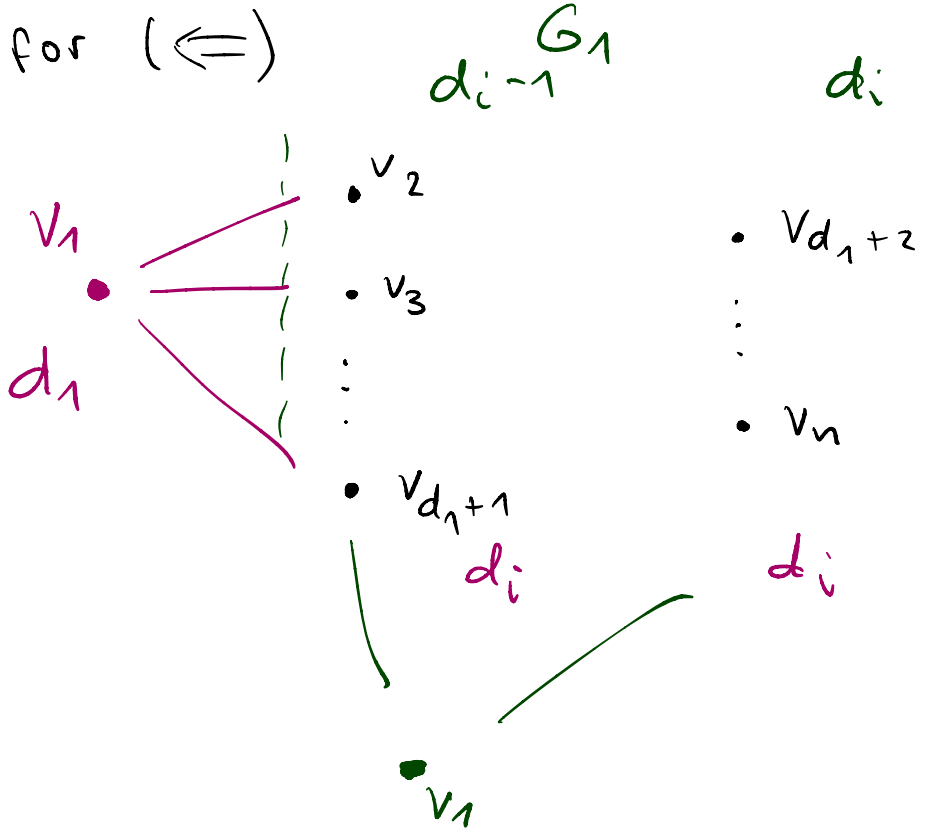
$$\deg v_i = \begin{cases} d_i - 1, & 2 \leq i \leq d_1 + 1 \\ d_i, & d_1 + 2 \leq i \leq n \end{cases}$$

Build G from G_1 by adding vertex v_1, $V(G) = \{v_1, v_2, \ldots, v_n\}$.
adding edges $v_1 v_i$ for each $2 \leq i \leq d_1 + 1$

Now $\deg_G v_i = \begin{cases} d_1 & i = 1 \\ \deg_{G_1} v_i + 1 & 2 \leq i \leq d_1 + 1 \\ \deg_{G_1} v_i & d_1 + 2 \leq i \leq n \end{cases} = d_i \quad \forall i.$

Thus s is graphical.

Sketch for ($\Leftarrow$)



$\Rightarrow$) Assume S is graphical.

Step 1: If S is realized by a graph H with $V(H) = \{v_1, \dots, v_n\}$, $\deg v_i = d_i$

such that $N_H(v_1) = \{v_2, v_3, \dots, v_{d_1+1}\}$, then we are done.

Because $G := H \setminus \{v_1\}$ will realize S_1 ✓

Step 2: There exists some H as in Step 1.

There are realizations we need to build a special one.

Take a realization G with $V(G) = \{v_1, \dots, v_n\}$, $\deg v_i = d_i$

for which $\sum_{v \in N_G(v_1)} \deg v$ is maximal.

Suppose $N_G(v_1)$ has d_1 elements but $\{\deg v : v \in N_G(v_1)\} \neq \{d_2, \dots, d_{d_1+1}\}$

There are some vertices v_m, v_M with $\deg v_m < \deg v_M$

but $v_m \in N(v_1) \not\ni v_M$

Since $\deg v_m < \deg v_M$, There is some v_t with $v_M v_t \in E(G)$ but $v_m v_t \notin E(G)$.

Build G' from G by erasing $v_1 v_m$, $v_M v_t$ and adding $v_1 v_M$, $v_m v_t$

Then G' has the same degree sequence as G

but now $$\sum_{v \in N_{G'}(v_1)} \deg v > \sum_{v \in N_G(v_1)} \deg v$$

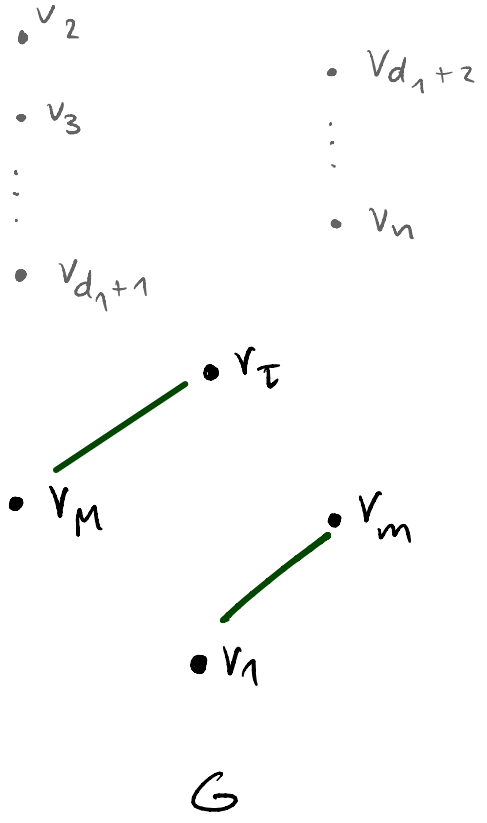
$$\cdots + \deg v_M + \cdots > \cdots + \deg v_m + \cdots$$

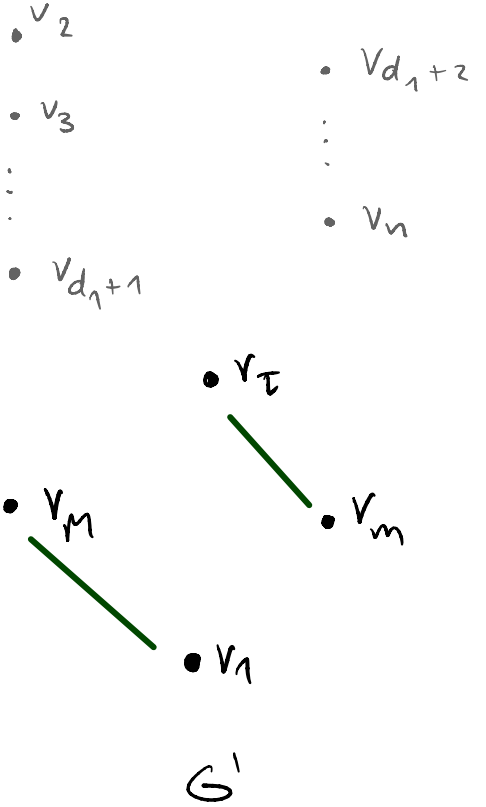
That contradicts G is maximal

Thus $\{\deg v : v \in N_G(v_1)\} = \{d_2, \ldots, d_{d_1+1}\}$

G as in step 1 !!

Sketch for ($\Rightarrow$)





Example (2.11). *Decide whether* s: 5, 4, 3, 3, 2, 2, 2, 1, 1, 1 *is graphical.*

s_1: 4-1, 3-1, 3-1, 2-1, 2-1, 2, 1, 1, 1

s_1: 3, 2, 2, 1, 1, 2, 1, 1, 1

s_1: ~~3~~, 2, 2, 2, 1, 1, 1, 1, 1

s_2: 2-1, 2-1, 2-1, 1, 1, 1, 1, 1

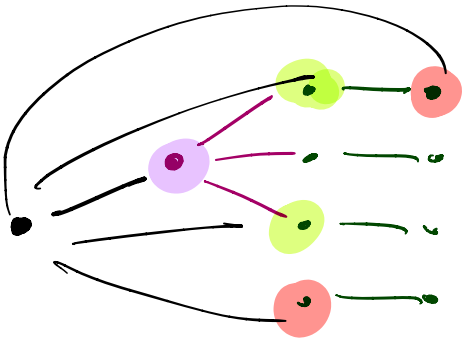
s_2: 1, 1, 1, 1, 1, 1, 1, 1

$s \rightsquigarrow s_1 \rightsquigarrow s_2 \rightsquigarrow$

$4, 3, 3, \dots, 1, (-1)$

$\Rightarrow$ not graphical.





Theorem (Erdös-Gallai (not in the book)). *A sequence* $s\colon d_1 \geqslant d_2 \geqslant \cdots \geqslant d_n$ *is graphical if and only if*

(1) $\sum_{i=1}^{n} d_i$ *is even, and* (2) *for all* $k \in \{1,\dots,n\}$, $\sum_{i=1}^{k} d_i \leqslant k(k-1) + \sum_{i=k+1}^{n} \min\{k, d_i\}$.

Proof. (Only $\Longrightarrow$) Assume $G = (V, E)$ is a realization of s with vertices $V = \{v_1,\dots,v_n\}$ such that $d_i = \deg v_i$.

Step 0: $\sum_{i=1}^{n} d_i$ is even: $\sum_{i=1}^{n} d_i = \sum_{v\in V} \deg v = 2\,|E(G)|$ even!

Now we prove (2) in a series of steps. Fix k and consider $\Omega = \{(x,y) \in \{v_1,\dots,v_k\} \times V\colon xy \in E(G)\}$.

Step 1: $|\Omega| = \sum_{i=1}^{k} d_i$:

Simply because $|\Omega| = \sum_{i=1}^{k} |\{y\in V : v_i y \in E(G)\}| = \sum_{i=1}^{k} \deg v_i = \sum_{i=1}^{k} d_i$

Step 2: We have a partition $\Omega = \Omega_1 \sqcup \Omega_2$, where

$\Omega_1 = \{(x,y) \in \{v_1,\dots,v_k\} \times \{v_1,\dots,v_k\}\colon xy \in E(G)\}$, $\Omega_2 = \{(x,y) \in \{v_1,\dots,v_k\} \times \{v_{k+1},\dots,v_n\}\colon xy \in E(G)\}$.

This is clear: an arbitrary element of Ω is of the form (x,y) where $xy \in E(G)$, $x \in \{v_1,\dots,v_k\}$ and y belongs to exactly one of $\{v_1,\dots,v_k\}$ or $\{v_{k+1},\dots,v_n\}$.

Step 3: $|\Omega_1| \leqslant k(k-1)$

$$|\Omega_1| \leq \left|\{(x,y) \in \{v_1,\dots,v_k\} \times \{v_1,\dots,v_k\} : x \neq y\}\right| = k(k-1)$$

Possibilities for x

Possibilities for $y \neq x$.

Step 4: $|\Omega_2| \leqslant \sum_{i=k+1}^{n} \min\{k, d_i\}$.

$$|\Omega_2| = \sum_{i=k+1}^{n} \left|\{x \in \{v_1,\dots,v_k\} : xv_i \in E(G)\right| \leq \sum_{i=k+1}^{n} |\{v_1,\dots,v_k\} \cap N(v_i)| \leq \sum_{i=k+1}^{n} \min\{k, \underbrace{\deg v_i}_{=d_i}\}$$

Step 5: Finish the proof of (2).

$$\sum_{i=1}^{n} d_i = |\Omega| = |\Omega_1| + |\Omega_2| \leq k(k-1) + \sum_{i=k+1}^{n} \min\{k, d_i\}$$

STEP 1 STEP 2 STEPS 3,4.

□